

GIADA SOSPIRO

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PROFESSIONAL SUMMARY

Biotechnologist with hands-on experience in molecular diagnostics, assay development, and scientific communication within GMP-regulated R&D environments. Skilled in qPCR optimization, analytical validation, and documentation aligned with regulatory and quality standards. Adept at translating experimental findings into clear technical reports, presentations, and data-driven insights that support decision-making in diagnostics and biotechnology projects.

CORE COMPETENCIES

Molecular Diagnostics • qPCR Assay Development • Analytical Validation • Experimental Design • Technical Reporting • Scientific Writing • Literature Review • Data Interpretation • Workflow Optimization • GMP/SOP Compliance • Cross-functional Communication • Python Automation • CRISPR-Cas9 • Cell Culture • Flow Cytometry • Scientific Presentations • Troubleshooting

PROFESSIONAL EXPERIENCE

R&D Scientist — Alifax R&D | Trieste, Italy

Oct 2022 – Nov 2025 - Drove development and refinement of qPCR assays for bloodstream pathogen detection, strengthening diagnostic performance and supporting device evolution. - Executed analytical validation workflows—including sensitivity, specificity, linearity, and reproducibility—to ensure robust, quality-controlled assay outputs. - Conducted literature evaluations to identify emerging biomarkers and assess feasibility of adding new molecular targets to the diagnostic pipeline. - Implemented Python-based automation to streamline reporting and reduce manual, error-prone documentation steps, improving workflow reliability. - Authored technical reports, data summaries, and scientific presentations to communicate findings to internal teams and external collaborators. - Upheld GMP and SOP requirements across all laboratory procedures, ensuring traceability, compliance, and data integrity.

Research Fellow — ICGEB | Trieste, Italy

Mar 2022 – Oct 2022 - Performed molecular and cellular experiments within a Telethon-funded immunology project to investigate mechanisms underlying Wiskott-Aldrich syndrome. - Conducted mouse handling, tissue collection, and primary cell isolation to support downstream functional assays. - Utilized CRISPR-Cas9 gene editing through both lentiviral delivery and nucleofection to explore BCAP-WASp pathway interactions. - Generated reliable datasets and maintained detailed, publication-ready experimental documentation. - Supported troubleshooting of complex protocols, enhancing reproducibility and workflow efficiency within the research group.

EDUCATION

Master's Degree in Medical and Pharmaceutical Biotechnology

University of Trieste | 2019–2022 | 110/110 cum laude Thesis: *"Priming cells to type I IFN production: is there a complex interplay involving BCAP and WASp?"*

- Designed gene-editing workflows and conducted immunological assays to study interferon signaling pathways.

Bachelor's Degree in Biotechnology

University of Perugia | 2016–2019 | 107/110 Thesis: *"Oral mucositis as a debilitating complication in chemotherapy protocols: the microbiota as a new treatment option."*

- Generated murine organoids to study the impact of microbiota-derived metabolites on cell renewal and differentiation.

TECHNICAL SKILLS

Molecular & Cellular Techniques:

qPCR, PCR, nucleic acid extraction, cDNA synthesis, cloning, bacterial culture, ELISA, Western blotting, CRISPR-Cas9 (lentiviral + nucleofection), mammalian and primary cell culture, murine handling, tissue dissociation.

Diagnostics & Laboratory Operations:

Assay development, analytical validation, sample processing, troubleshooting, documentation, quality control, GMP/SOP compliance.

Data & Digital Tools:

Python (workflow automation), GraphPad Prism, FlowJo (basic), Microsoft Excel/Word/PowerPoint/Publisher, BioRender.

Communication Skills:

Scientific writing, technical reporting, literature review, data visualization, presentation preparation.

Languages:

Italian (native), English (B2 professional proficiency).

PUBLICATION

"BCAP Is an Interferon-Stimulated Gene That Enhances Type I Interferon Activity in Response to Lipopolysaccharide."

Contributed experimental design, molecular assays, and data generation supporting the study.